

Acceptance Test Plan

for

Race Walking Data Visualizer

Version 1.0

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1. Document History

Name	Date	Reason	Version
Elvis Cao, Ostap Bezrukyy, Jiacong Li, Sotero Casiano, Nico Saputra, Jay Dave, Zach Bridgers	April 20, 2025	Initial Draft	1.0

2. Introduction

2.1 Purpose

This document provides the plan for completing the acceptance test of the Race Walking Data Visualizer. The included test cases were derived from the Software Requirement Specifications for Race Walking Data Visualizer (2.2.1), which details the system's functionality.

2.2 References

2.2.1 Software Requirements Specification

This document references the requirements in the Software Requirements Specification for Race Walking Data Visualizer document. Requirement numbers from this document are listed for every test to indicate the requirements the test exercises.

2.3 Definitions

Bent Knee: Angle of an athlete's leg when the foot is in contact with the ground and the leg's position is before it reaches the vertical position. This infraction symbol is denoted by <

Loss of Contact (LOC): Amount of time when both athlete's feet are not in contact with the ground. This infraction symbol is denoted by ~

Python3: A high-level programming language utilized for driving data pulls, data organization, and visualization of graphs within the Race Walking Data Visualizer.

3. Test Approach and Constraints

3.1 Test Objectives

Following the prescribed Acceptance Test Plan process verifies Race Walking Data Visualizer fulfills all of the functional requirements specified in the Software Requirements Specification. Race Walking Data Visualizer is ready for deployment if and only if it passes all of the test cases contained in the Acceptance Test Plan

3.2 Test Structures

Atomic test cases were derived from the SRS. Each test case has a unique identifier, a name, a high-level description of the functionality it tests and references to its related requirements in the SRS. Additionally, each test case contains preconditions, actions and postconditions.

Preconditions describe the required state of the system before the test is performed. The actions include an itemized list of steps the tester takes when performing the test. The postconditions describe the state the system is expected to be in after the actions are performed. If the postcondition(s) are met, then the software passes the test case.

3.3 Test Constraints

The Acceptance Test Plan only tests the functionality of the Race Walking Data Visualizer as described in the SRS. The test cases described in the Acceptance Test Plan are not concerned with specific design and implementation details of the system. The test cases focus on ensuring the correct system behavior is achieved for the complete set of functions available to users of the system

4. Test Assumptions and Exclusions

This section provides greater details about which functions and features of the Race Walking Data Visualizer and are not covered by the Acceptance Test Plan process.

4.1 Test Assumptions

The test cases covered in the Acceptance Test Plan are written under the assumption that the related issues are also addressed by unit tests, integration tests and system tests for the Race Walking Data Visualizer.

The Acceptance Test Plan covers:

- The functional requirements listed in the Software Requirements Specification.
- The consistency of the user-related system documentation.
- The usability of the system.

4.2 Test Exclusions

It is assumed that unit tests, integration tests, and system tests for the Race Walking Data Visualizer exercise areas of the system not covered by the Acceptance Test Plan.

The Acceptance Test Plan does not cover:

- The non-functional requirements of the system listed in the Software Requirements Specification.
- Functionality beyond what is described in the Software Requirements Specification
- The structural integrity of the source code.

5. Entry and Exit Criteria

This section lists the criteria that must be satisfied in order for the Acceptance Test Plan to commence, as well as the criteria that must be satisfied in order to conclude the acceptance testing.

5.1 Entry Criteria

Acceptance testing begins after the following preconditions are met:

- All Priority 1 requirements are considered to be implemented by the Race Walking Data Visualizer development team.
- Unit, integration and system testing for the Race Walking Data Visualizer are completed successfully.
- A proper testing environment meeting the hardware and software constraints specified in 4.3 of the Software Requirements Specification is available.
- The Test Team Leader and the Stakeholder received the most recent revision of the Software Requirements Specification.
- Consent is obtained from the Test Team Leader.
- Consent is obtained from the Stakeholder.

5.2 Exit Criteria

The Acceptance Test Plan can end after any of the following conditions are met:

- All Priority 1 requirements were tested and resulted in expected behavior (Success).
- At least one Priority 1 requirement failed to meet the documented specification (Failure).
- All parties mutually agree to postpone testing until a later date (Failure).

6. Testing Participants

This section describes the roles and responsibilities of the parties involved in the Acceptance Test Plan, as well as the procedure for reporting the test results and any subsequent issues.

6.1 Roles and Responsibilities

For the Acceptance Test Plan, the following individuals fill the roles listed below:

- Test Team Leader: Elvis Cao
- Testers: Ostap Bezrukyy, Sotero Casiano, Jiacong Li, Nico Saputra, Jay Dave, Zach Bridgers
- Stakeholder: Jeff Salvage

6.2 Training Requirements

All parties involved in the Acceptance Test Plan should be comfortable with standard web application interfaces. Additionally, all parties should be familiar with the Race Walking Data Visualizer user interface, the type of visualizations Race Walking Data Visualizer is expected to produce, and the Race Walking Data Visualizer Software Requirements Specification

6.3 Problem Reporting

Any problem found by either a Tester or the Stakeholder must be documented and reported to the Test Team Leader via the project's issue tracking system. Any reported problems are to be reviewed by the Test Team Leader and development team during a mandatory post testing review meeting and subsequently assigned to at least one person on the development team to resolve.

6.4 Progress Reporting

The Acceptance Test Plan Report is to be compiled by the Test Team Leader after the testing process has been concluded and has satisfied any one of the conditions listed in 5.2. The Race Walking Data Visualizer development team is to review the Acceptance Test Plan Report during the mandatory post testing review meeting.

7. Test Cases

7.1 Introduction

The test cases below cover all functionality required by the Software Requirements Specification document. Each of the test cases includes the following information:

- ID - An identification code for the test case
- Name - A descriptive name of the test
- Requirement(s) - The requirement number(s) from the Software Requirements Specification document
- Description - A brief overview of the test purpose
- Precondition(s) - The expected state of the software before the actions are executed
- Action(s) - The step(s) to be completed by the tester
- Postcondition(s) - The expected state of the software after the actions are executed

7.2 Test Cases

7.2.1 Launching the Race Walking Data Visualizer

ID	T7.2.1.1
Name	Launching the Race Walking Data Visualizer
Requirement	R3.0.1, R3.0.2, R3.0.3, R3.0.4
Description	The user launches the Race Walking Data Visualizer application and views the main menu page
Precondition	The user has cloned the Race Walking Data Visualizer repository to the user's local computer and installed the requirements.txt file.
Action	1. The user opens a web browser by running app.py from the terminal
Postcondition	• The user is presented with the main menu page • The main menu page consists of the Data Graphs, Report Generator, Web Scraper, Task Runner, and Data Modifier links

7.2.2 Data Modifier

ID	T7.2.2.1
Name	Viewing Database Tables
Requirement	R3.1.1, R3.1.2.1, R3.1.2.3, R3.1.2.4, R3.1.2.5, R3.1.2.6
Description	The user can see all database tables, navigate them, and search for entries
Precondition	The user has launched the Race Walking Data Visualizer and navigated to the Data Modifier page.
Action	1. The user clicks 'Athlete' in the navbar to jump to the athlete table. 2. The user enters 'ESP' in the search bar. 3. The user hits 'next' to go to the second page of results. 4. The user clicks the 'LastName' column header to order the column by last name.
Postcondition	• The user can view the second page of the athlete table, sorted by last name, and filtered for athletes from Spain.

ID	T7.2.2.2
Name	Editing Database Tables
Requirement	R3.1.2.2
Description	The user can edit individual cells in the database tables
Precondition	The user has launched the Race Walking Data Visualizer and navigated to the Data Modifier page.
Action	1. The user clicks 'Athlete' in the navbar to jump to the athlete table. 2. The user clicks on the BibNumber of the first athlete in the table. 3. The user enters a new value and clicks off the cell.
Postcondition	• The user can reload the page and see the edit they made in the table.

ID	T7.2.2.3
Name	Deleting/Creating rows
Requirement	R3.1.2.2
Description	The user can delete and create rows in the database tables
Precondition	The user has launched the Race Walking Data Visualizer and navigated to the Data Modifier page.
Action	1. The user clicks either “delete” or “add row” button in the actions column 2. If deleting, the user clicks OK to proceed
Postcondition	• The user can reload the page and see the changes they have made

7.2.3 Data Graphs

ID	T7.2.3.1
Name	Generate athlete graph
Requirement	R3.2, R3.3
Description	The user can generate a graph to visualize up to 10 athletes to view their infractions, judge calls, and finishing times
Precondition	The user has launched the Race Walking Data Visualizer and navigated to the Data Graphs page.
Action	1. The user highlights up to 10 athletes in the left box 2. The user moves those athletes to the right box by clicking the right arrow 3. When done, the user clicks the Generate Graph button to generate a graph showing athlete data on infractions, judge calls, and finishing times
Postcondition	• The user can view the athlete graph that they generated above • They can differentiate between athletes based on their color scheme in the athlete legend • They can differentiate between judges based on the judge legend containing the judge #s • They can view infractions such as B, ~, or < • They can view whether the infraction was a red or yellow card

	• They can view when an athlete has finished the race
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7.2.4 Report Generation

ID	T7.2.4.1
Name	Generates and downloads PDF for every report
Requirement	R3.4.3
Description	For the following reports the Download PDF button generates and downloads a PDF including the report table: 1. All Races 2. Athlete Infractions 3. Average Bent Knee By Athlete Across All Observations 4. Average LOC By Athlete Across 2 Worst Observations 5. Consistency Discrepancies 6. Individual Judge Calls Vs Video Observations 7. Infraction Summary By Judge 8. Judge Consistency Discrepancy 9. Judge Consistency Missed 10. Legal Athletes With 3 Or More Red Cards 11. List All Judges In A Race 12. Majority Matched 13. Red Cards Without Yellow Paddle 14. Total Score 15. Total Score 55 16. Total Score Extra Column 17. Video Observations Infraction Totals 18. Video Observations Infractions Static Panned 19. Video Observations Infractions Total 20. Video Observations Panned Infractions Total 21. Yellow Paddles Without Red Cards
Precondition	The user has launched the Race Walking Data Visualizer and navigated to the Report generator page.
Action	1. The user selects a report from the dropdown, the default styling, the Budapest race, and the Complete database as a representative sample report. 2. The user then hits the Download pdf button. 3. Repeat for all reports.
Postcondition	• The report is generated on the web page immediately upon being selected.

	• When the Download PDF button is pressed, a report is generated that is identical to the one displayed on the web page. • The margins, styling and data of the report is all correct. • The file downloads and opens up in a new tab.
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ID	T7.2.4.2
Name	Proper error handling for missing data
Requirement	R3.6
Description	Reports incompatible with the chosen databases alert the user to the invalid nature of the request cleanly and clearly.
Precondition	The user has launched the Race Walking Data Visualizer and navigated to the Report generator page.
Action	1. The user attempts to run the All Races query on the Drexel Race walking database, by selecting the All Races query and the Drexel Race Walking database from the dropdown menus.
Postcondition	• The web page displays in place of a table report result, displays a string telling of what went wrong beneath the form.

7.2.5 Web Scraping

ID	T7.2.5.1
Name	Scraping athlete data using the Web Scraper
Requirement	R3.5.1, R3.5.2, R3.5.3
Description	The user scrapes athlete data from a supported external website (e.g., World Athletics) using the Web Scraper functionality. The scraped data is stored locally in a SQLite database.
Precondition	• The user has launched the Race Walking Data Visualizer. • The user has navigated to the Web Scraper page from the main menu. • An active internet connection is available.
Action	1. The user enters a valid athlete name into the Web Scraper search bar. 2. The user clicks the button. 3. The website launches a headless browser session using Selenium. 4. The scraper navigates to the athlete's page and extracts relevant data (e.g., name, country, race results, infractions). 5. The scraper saves the extracted data to scraped_data.db.
Postcondition	• The new athlete record is added to the scraped_data.db database • Scraped data appears on the website

